Code No: 158DP **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, July/August - 2022

MEASURING INSTRUMENTS

(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss about the statistical analysis of measurement data with an example.
b) A 0-150V voltmeter has a guaranteed accuracy of 1 percent of full scale reading. The voltage measured by instruments is 75V. Calculate the limiting error in percent. [8+7]

2.a) Explain in detail about Resistance Standards.
b) Discuss about the different forms and methods of measurements. [7+8]

3.a) With a neat diagram, explain the operation of Resistive Temperature Detectors.
b) Describe the working of Eddy current sensors and this application. [8+7]

4.a) What is a Potentiometer? Describe the working of Potentiometer with a neat sketch.
b) Explain the working of a Differential capacitor with a neat diagram. [8+7]

5.a) Discuss the design of Linear Measurement Instruments in detail.
b) Define a comparator. Discuss the functional requirements of a comparator. [8+7]

6.a) Describe the measurement of Rotational Velocity with a neat diagram.
b) List the types of Accelerometers. Explain any one type of accelerometers in detail.[7+8]

7.a) Define Manometer. Briefly explain about inverted U-Tube Manometers in detail?
b) With a neat sketch, explain the principle of Vibrating wire Force Transducer. [8+7]

8.a) Describe with neat sketch, the construction and working Variable area Flow meter.
b) With a neat diagram explain the operation of two float viscometer. [8+7]

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